

KARNATAKA BOARD

CLASS 10

SCIENCE

2016

Max. Marks: 80

i) The Question-cum-Answer Booklet consists of objective and subjective types of questions having 42 (questions.

ii) Space has been provided against each objective type question. You have to choose the correct choice and write the complete answer along with its letter in the space provided.

iii) For subjective type questions enough space for each question has been provided. You have to answer the questions in the space.

iv) Candidate should not write the answer with pencil. Answers written in pencil will not be evaluated. (Except Graphs, Diagrams & Maps)

v) Answer only one question each for the choice questions.

vi) Follow the instructions given against both the objective and subjective types of questions.

vii) In case of Multiple Choice, Fill in the blanks and matching questions, scratching / rewriting / marking.

viii) Space for Rough Work has been printed and provided at the bottom of each page.

ix) Candidates have is not permitted, thereby rendering to disqualification for evaluation extra 15 minutes for reading the question paper.

x) Do not write anything in the space provided in the right side margin.

Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter in the space provided against each question. $10 \times 1 = 10$

- 1. In Fleming's right hand rule middle finger indicates the direction of**
(A) Magnetic field (B) induced electric current
(C) Mechanical energy (D) motion of the conductor.

Ans: (B) induced electric current

2. Identify one of the uses of solar heater in the following.

(A) Conversion of solar energy into electrical energy (B) Providing energy for artificial satellites.

(C) Desalination of marine water (D) Using in automatic streetlights.

Ans: (A) Conversion of solar energy into electrical energy

3. Significant reduction in the platelet count of human blood is the main symptom of this disease. (A) Chikungunya (B) Dengue

(C) Bird flu (D) Gonorrhea.

Ans: (B) Dengue

4. $\text{Si} + \text{C} \rightarrow \text{SiC}$. This chemical reaction is an example for

(A) Exothermic reaction

(B) endothermic reaction

(C) Dissociation reaction

(D) displacement reaction.

Ans: (A) Exothermic reaction

5. No image is formed on the blind spot of human eye because

(A) Cones are absent

(B) rods are absent

(C) Rods and cones are absent (D) optic nerve is absent.

Ans: (C) rods and cones are absent

6. A domestic electrical appliance requires alternating current of 15 V. If 220 V of alternating current is supplied to the house, then the device that helps in the functioning of that electrical appliance is (A) induction coil (B) step-up transformer

(C) AC dynamo

(D) step-down transformer.

Ans: (B) step-up transformer

7. While doing the experiment of coppervoltameter, in which of the following cases, more amount of copper gets deposited at cathode?

(A) 2 amperes of electric current is passed for 30 minutes

(B) 4 amperes of electric current is passed for 20 minutes

(C) 0.5 amperes of electric current is passed for 80 minutes

(D) 1.5 amperes of electric current is passed for 30 minutes.

Ans: (B) 4 amperes of electric current is passed for 20 minutes

8. Identify the correct complementary base pairing among the following.

- (A) Adenine — Thymine and Guanine — Cytosine
- (B) Adenine — Guanine and Thymine — Cytosine
- (C) Adenine — Cytosine and Thymine — Guanine
- (D) Guanine — Adenine and Cytosine — Adenine.

.Ans: (A) Adenine — Thymine and Guanine — Cytosine

9. Find out the most efficient engine in the following.

- (A) An engine converts 80 KJ of heat energy into 20 KJ of work
- (B) An engine converts 50 KJ of heat energy into 15 KJ of work
- (C) An engine converts 30 KJ of heat energy into 6 KJ of work
- (D) An engine converts 60 KJ of heat energy into 24 KJ of work

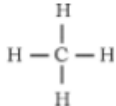
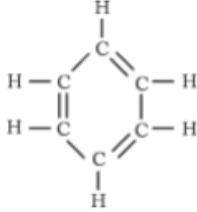
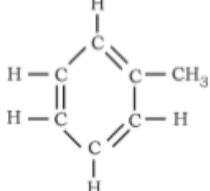
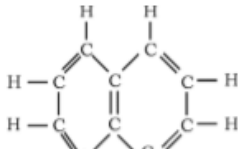
Ans: (D) an engine converts 60 KJ of heat energy into 24 KJ of work

10. Haemoglobin levels in the blood samples of two persons A and B are found to be 9 gm/dL and 13 gm/dL respectively. Which statement is true with respect to the amount of oxygen supply in their body?

- (A) More in person B than in person A
- (B) More in person A than in person B
- (C) Equal in person A and person B
- (D) No correlation between oxygen supply and the level of haemoglobin.

Ans:(A)More in person B than in person A

11. The structural formulae of hydrocarbons are given in Column-A and their uses are given in Column-B. Match them and write the answers along with its letters in the space provided: $4 \times 1 = 4$

Column - A	Column - B
(A) 	(i) preparation of moth balls
(B) 	(ii) solvent for lacquers (iii) preparation of soap (iv) dry cleaning
(C) 	(v) used as fuel (vi) preparation of esters
(D) 	(vii) preparation of aspirin.

Ans:

- (A) Used as a fuel
(B) Dry cleaning
(C) Preparation of Aspirin
(D) Preparation of moth balls

Answer the following questions:

$$7 \times 1 = 7$$

12. Write any two advantages of bio-energy.

Ans: Two advantages of bio-energy:

- It maintains and polluted environment.
- IT reduces the carbon dioxide content in the atmosphere.

13. What is red shift?

Red shift is the displacement of the spectrum of an astronomical object toward longer (red) wavelengths. It is generally attributed to the Doppler Effect.

14. Name the two important non-metallic oxides which cause acid rain.

Ans: Sulphur oxides and nitrogen oxides

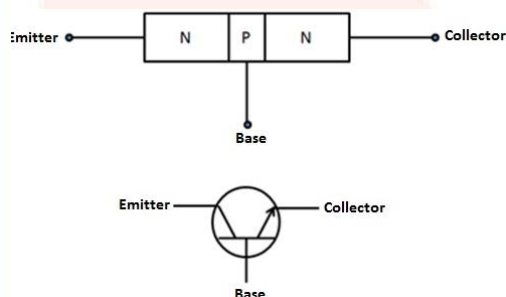
15. Mention the properties of silica due to which it is used as sand bath in laboratory.

Ans: Fused silica and natural sand have similar behavior. Especially for stress-strain behavior. Fused silica can be used to model natural sand under a low confining pressure condition. An appreciable breakage is noticed in fused silica once the confining stress exceeds 300 kPa

16. State the modern periodic law.

Ans: This law states that Physical and chemical properties of the element are periodic function of their atomic number.

17. Draw the circuit symbol of n-p-n transistor.



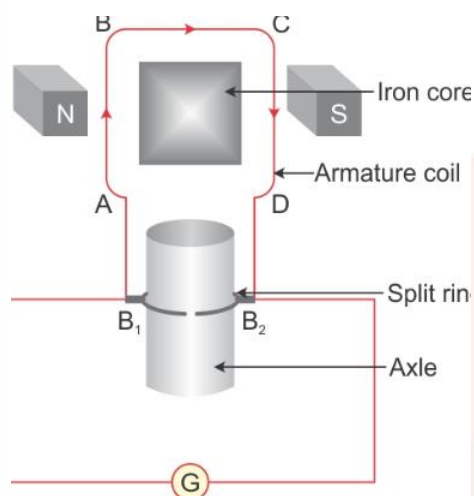
18. Sclerenchyma fibres are used in coir industries to make gunny bags and ropes. Give scientific reason.

Ans: Sclerenchyma fibres are flexible and elastic hence they are used in coir industries to make gunny bags and ropes.

Answer the following questions:

$16 \times 2 = 32$

19. Draw the diagram of a DC dynamo and label the following parts.



(a) Split rings (b) Armature coil.

20. What is annealing of glass?

Ans: Annealing of glass is a process of slowly cooling hot glass objects after they have been formed, to relieve residual internal stresses introduced during manufacture.

Name the colour obtained when

(i) Ferric compounds, (ii) Cobalt compounds are added to the mixture of molten glass.

Ans: The ferrous oxide imparts a green colour to glass and ferric oxide imparts a very light yellow tint.

Cobalt compounds imparts dark blue colour.

OR

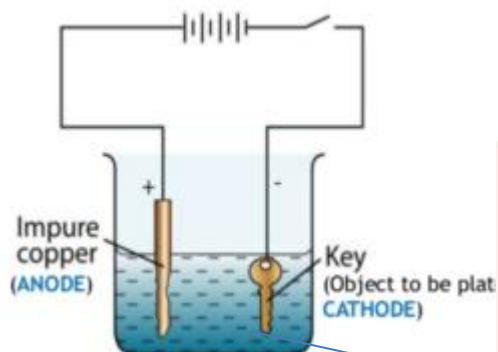
What is pulping? How is an uncoated paper converted into coated paper?

Ans: The process of making pulp, especially from wood, for use in the manufacture of paper, cardboard, etc. is called pulping. Coated papers contain a layer of coating material on one or both side. The coating materials of pigments and fillers help to improve the printing surface of the paper.

21. Draw the diagram of the apparatus used in electroplating and label the following parts.

(i) Electrolyte

(ii) Anode.



Electrolyte

22. Lymph plays an important role in protecting immune system of the body. Justify this statement.

lymph is the fluid that removes cellular waste products, pathogens, and dead blood cells from the tissues. Lymphocytes present in lymph destroy the harmful pathogens.

OR

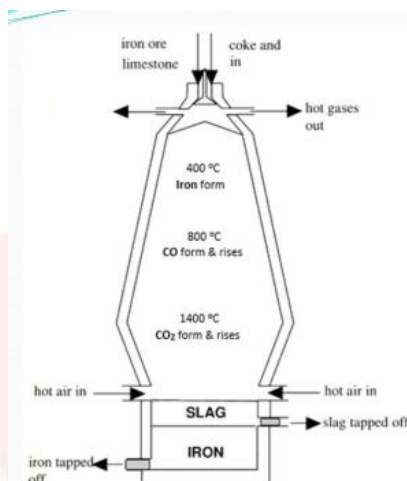
(a) How is dermal tissue adapted to prevent excessive transpiration in plants?

The epidermis is a dermal tissue that is usually a single layer of cells overlaying the more youthful elements of a plant. It secretes a waxy layer known as the cuticle that inhibits water loss.

(b) Why do leaves of lotus plant float on water?

Ans: leaves of lotus plant float on water because upward force generated by the fluid (generally water) opposes the downward force exerted by the weight of the lotus. The leaves also happen to be very broad in comparison to their weight meaning very little pressure is exerted on any part of the water. Their waxy leaves stop them taking on water, and helps repel any water attempting to gather on top.

23. Draw the diagram of blast furnace used in the extraction of iron.



24. Differentiate between Caucasoid man and Mongoloid man based on their physical features.

Ans: The Caucasoid have pale reddish white to olive brown skin colour. Among the Caucasoids, head is generally long to broad and short and medium high to very high.

Among the Mongoloids, the skin colour ranges between saffron to yellow brown, while some individuals have reddish brown skin colour. The Mongoloid's head is predominantly broad and medium high.

25. What is forward biasing and reverse biasing of a diode?

In electronics, biasing means applying D.C. Voltage to an electronics device to establish certain operating conditions.

A p-n junction is said to be forward biased when the external d.c voltage applied in such a direction that it cancels the potential barrier, hence, permitting the current flow. A pn junction is said to be reverse biased when the external d.c voltage applied to the junction is in such a direction that it increases the potential barrier.

OR

What are extrinsic semiconductors? Name the two types of extrinsic semiconductors.

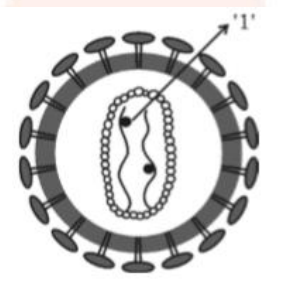
Ans: Extrinsic semiconductors are semiconductors that are doped with a specific impurity. The impurity, in turn, modifies the electrical properties of the semiconductor and makes it more suitable for use in electronic devices such as diodes and transistors. There are two types of extrinsic semiconductors.

N-type semiconductors and P-type semiconductors

26. Sodium (Na), Magnesium (Mg), Aluminium (Al) and Silicon (Si) are arranged in the decreasing order of their atomic size. Which element has the highest ionisation energy among them? Justify your answer scientifically.

Ans: Ionization energy is the energy required to remove an electron from an atom. The more attracted the electrons are to the nucleus, the harder it is to remove the electron. A higher atomic number, then, which corresponds to higher number of protons and electrons, would mean a higher ionization energy and hence the ionization energy increases along a period so magnesium has highest ionization energy.

27. Draw the diagram showing the structure of HIV.



28. The densities of 4 gases at standard temperature and pressure are given in the table:

Gas	Methane	Ammonia	Helium	Neon
Density	0.72 g/L	0.77 g/L	0.18 g/L	0.90 g/L

Among these gases, which gas diffuses very fast? State the law that helps you to take the decision.

Ans: Helium diffuses first and then methane, ammonia and neon in order. The extent to which the gases can diffuse mainly depends upon its density. Fick's laws of diffusion describe diffusion and were derived by Adolf Fick in 1855. They can be used to solve for the diffusion coefficient, D. Fick's first law can be used to derive his second law which in turn is identical to the diffusion equation.

29. Hydroponic and aeroponic methods are gaining significance in space research organisations. Give two reasons for this.

Ans: Cultivation of plants by placing the roots in the nutrient solution without any soil is called hydroponics. It is also known as soilless culture/ water culture/ solution culture. When the plants are grown without soil in an artificial medium, the technique is called as aeroponics. This allows the plants to be cultured and survive in the space when the plant is supplied with proper nutrients. The plants can grow and survive without soil in any place. Thus, the research on the plants can be performed in space and on the surface of the other planets.

OR

Urban people should be encouraged to take up the practice of roof-top gardening. Justify.

Ans: A roof garden is a garden on the roof of a building. Besides the decorative benefit, roof plantings may provide food, temperature control, hydrological benefits, architectural enhancement, habitats or corridors for wildlife, recreational opportunities, and in large scale it may even have ecological benefits.

30. The practice of cultivating food on the rooftop of buildings is sometimes referred to as rooftop farming.30. What are ultrasonic waves? Write any two uses of ultrasonic waves in the field of medicine.

Ans: Ultrasonic waves are vibrations of frequencies greater than the upper limit of the audible range for humans—that is, greater than about 20 kilohertz. The term sonic is applied to ultrasound waves of very high amplitudes.

Two uses are followings:

Ultrasonic sound is used in ultrasound scan to see the progress of the baby inside the mother's womb. Ultrasonic waves are used to break any stone which is present in any of the body organ.

OR

What is an echo? Name the two devices which work on the principle of echo of ultrasonic waves

Echo is a repetition of sound produced by the reflection of sound waves from a wall, mountain, or other obstructing surface. The two devices which work on the principle of echo of ultrasonic waves are

- SONAR (sound navigation and ranging) and ultrasonic sonography.

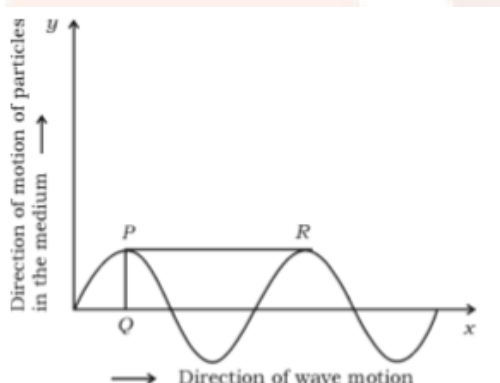
31. Hydroelectric power plants are more ecofriendly than thermal power plants. Justify this statement.

Ans: Hydropower is fueled by water, so it's a clean fuel source, meaning it won't pollute the air like thermal power plants coal or natural gas. Hydroelectric power is a domestic source of energy, allowing each state to produce their own energy without being reliant on international fuel sources.

32. The general formula of a group of organic compounds is $C_n H_{2n+1} OH$. Write the molecular formula of first two members of this group. Examine whether these two compounds are in homologous series, based on their molecular formula.

Ans: The first member is Methanol. Molecular formula is CH_3OH . Second member is Ethanol. Molecular formula is C_2H_5OH . They are in homologous series because they have same function group.

33. The graph of a wave motion is given below. Observe the graph and answer the following questions:



(a) What type of wave is represented in the graph? (b) What do PQ and PR indicate with respect to the wave?

Ans: (a) Longitudinal wave is represented in the graph.

(b) PQ represent the amplitude and PR represent the wavelength

34. A student catches small aquatic creatures swimming in a pond and keeps them in his aquarium assuming them to be fishes. After a few days he observes that four limbs are developed in those creatures. Then,

(i) To which class of vertebrates do you include those aquatic creatures?

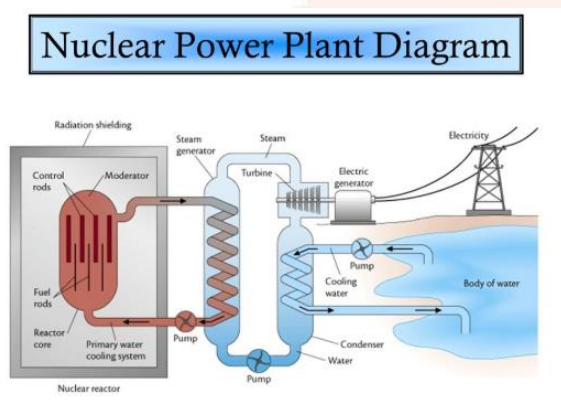
Ans: These vertebrates are included in amphibian.

(ii) Name the process that has caused development of four limbs in them.

Answer the following questions: $5 \times 3 = 15$

35. Draw the diagram of a nuclear power plant and label the following parts.

(a) Control rods (b) Radiation shield.



36. Name the steps in the manufacture of common sugar from sugarcane and explain the first step.

Ans: The following points highlight the six main steps involved in manufacturing cane-sugar. The steps are:

1. Extraction of the Juice
2. Clarification of Juice
3. Concentration and Crystallization
4. Separation of Crystals
5. Refining of Sugar
6. Recovery of Sugar Molass

Extraction of the Juice

The canes are thoroughly cleaned and cut into small pieces. These cane pieces are dropped over a continuous belt of steel plates called the cane-carrier, moving over roller chains. From the cane-carrier, these are conveyed to a crusher fitted with a set of revolving knives, and cut into very small pieces.

These small pieces in the form of a compact blanket are made to pass through two roller crushers and four sets of mills. After major quantity of the juice has been extracted by the

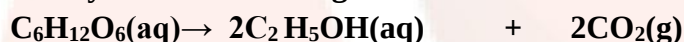
crushers and the first two mills, cold or hot water is sprinkled over the bagasse whereby residual juice gets diluted and can be easily extracted by further milling.

OR

(a) What is fermentation? Give one example.

Ans: Fermentation occurs when a microorganism, such as yeast or bacteria, converts carbohydrates into another chemical. Yogurt is made from fermented milk. Yogurt is created through the combined action of the bacteria *Streptococcus lactis* and *Lactobacillus bulgaricus* or *Lactobacillus acidophilus*. These bacteria consume the sugar in the milk called lactose, curdle the milk and create the chemicals lactic acid and acetaldehyde, which give yogurt its distinctive taste.

(b) Write the balanced chemical equations of the reactions taking place in the manufacture of ethyl alcohol from sugar.



37. (a) Bryophytes are used in pots. Why?

(b) Explain alternation of generations in Bryophytes.

Ans: The haploid phase (n) is the gametophyte or sexual generation. It bears the sexual reproductive organs which produce gametes, i.e., antherozoids and eggs. With the result of gametic union a zygote is formed which develops into a sporophyte. This is the diploid phase (2n). The sporophyte produces spores which always germinate to form gametophytes.

During the formation of spores, the spore mother cells divide meiotically and haploid spores are produced. The production of the spores is the beginning of the gametophytic or haploid phase. The spores germinate and produce gametophytic or haploid phase. The spores germinate and produce gametophytes which bear sex organs.

38. (a) Explain the expansion stroke of a petrol engine.

Ans: At the end of the compression stroke when the piston is at TDC a pre metered quantity of diesel is injected into the cylinder by the injector. The temperature inside the cylinder is very high which is sufficient to ignite the fuel injected and this generates tremendous energy which is in the form of high pressure which pushes down the piston. The connection rod carries this force to the crankshaft which turns to move the vehicle. At the end of power stroke the piston reaches the BDC.

(b) There is no spark plug in diesel engine. Why?

Ans: Diesel engines don't require any spark plug because the required heat to ignite the fuel is generated due to compression of air.

39. What is bio-technology? List any two advantages and two limitations of biotechnology.

Ans: The advantages of biotechnology include curing infectious diseases, creating more efficient fuels and increasing farming yields to feed more people. Disadvantages of biotechnology include antibiotic resistant bacteria, new allergic reactions and higher prices for farmers.

OR

(a) Why did Mendel choose pea plants for his experiments? Give any four reasons.

Ans: Mendel selected the green pea for his experiment due to following reason.

- It can be easily grown or in the garden flower pot.
- The lifespan of this plant is very short and it completes its life cycle in one year.
- Its many generations can be studied in in less time easily.
- Its flower are bisexual.

(b) State Mendel's law of independent assortment.

Ans: Mendel's law of independent assortment states that when two pairs of traits are combined in a hybrid segregation of one pair of character is independent of the other pair of characters.

Answer the following questions: $3 \times 4 = 12$

40. (a) Which stage is attained by the star after the steady state ? Explain that stage.

(b) State Hubble's law.

Ans: Hubble's Law states that the recessional velocity of a galaxy (V) is equal to the product of the Hubble's Constant (H) and the distance to that galaxy (D) or $V = HD$

(c) A satellite is to be launched from the surface of the earth. Name the factors on which the escape velocity of the satellite depends.

Ans: Factors on which the escape velocity of the satellite depends are Center of mass and radius.

OR

(a) “Multistage rockets reduce the fuel consumption.” How? Explain.

Ans: Multistage fuel system consists of various tanks mounted or fitted on rocket. These tanks are capable of storing thousands of liter of fuel and that increases size and weight of empty tank. If single stage fuel system is used then rocket has to carry the weight of tank throughout the journey and that means after a point of time dead weight increase that is not of any use. Instead of using single stage fuel system, it is better to use multi stage system in which fuel tanks are used stage by stage and empty tanks are being detached from rocket that decrease weight to be carried. And as we know lesser the weight higher the efficiency. Thus rockets are occupied with multistage fuel system

(b) Explain how a neutron star is formed.

Ans: Neutron star results from the death of massive stars, which are around 8-15 times more massive than our sun. When a massive star collapses and explodes in a tremendous supernova explosion, the dense inner core is all that remains.

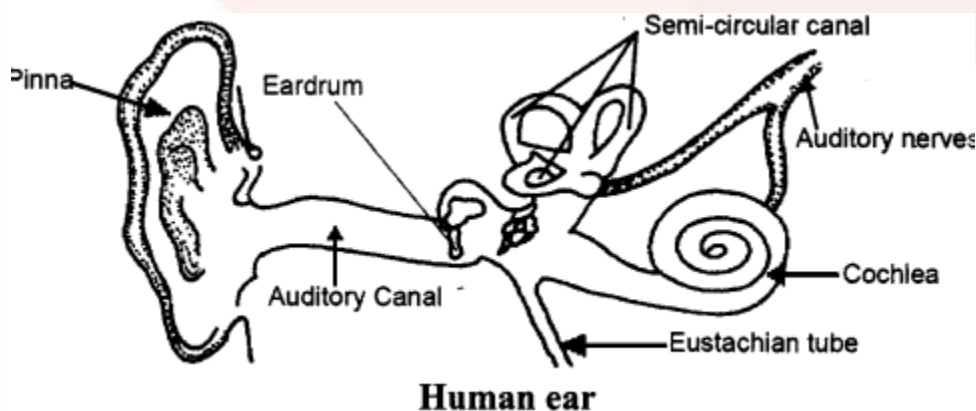
Ans: When the dying star is collapsing, the core gets progressively denser. Eventually, the elementary particles in the core (protons and electrons) cannot withstand the force of gravity and they fuse together to form neutron. Eventually all of the material in the core is converted into neutrons and the origin of the object name become obvious.

In which stage of the star, does nuclear fusion reaction begin?

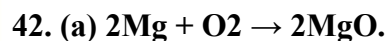
Ans: Fusion is the combining of atomic nuclei into larger atoms. The process releases huge amounts of energy. We achieve it currently only atomic bombs that use a fission nuclear reaction driver and at a very small and temporary scale with laser implosion of tritium.

In stars the driving force is the gravitational force of the stars mass. Thus, a star's nuclear fusion processes may be said to “start” when the accreting mass is sufficiently dense to force small nuclei together.

41. Draw the diagram showing the internal structure of human ear and label the following parts:



(a) Organ of Corti (b) Auditory nerve.



In this reaction, explain the experiment that you conduct to decide the product as a basic oxide.

Ans: Litmus Paper test is used to decide the product as a basic oxide. We Test magnesium oxide solution with litmus paper. The solution does not affect blue litmus but turns red litmus blue. This tells us that magnesium oxide is basic or alkaline in nature. In general, metal oxides are basic in nature.

(c) Name the method of concentration of Haematite ore and explain the method.

Ans: Haematite ore being oxide ore is concentrated by gravity separation process using hydraulic separator. The oxide is being heavier settles to the bottom while lighter impurities come to the surface and is removed.